Vivek Vishwanath

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Education

Georgia Institute of Technology Graduation Date: May 2028
B.S./M.S. Computer Science *Major GPA: 4.0*

- **Graduate Coursework**: Advanced Operating Systems, Advanced CPU Architecture, High-Performance Computing, Distributed Systems, Compiler Design
- **Undergraduate Coursework**: Computer Networking, Computer Architecture, Data Structures & Algorithms
- **Concentrations**: Systems & Architecture, Artificial Intelligence

Experience

Palantir May 2026 – August 2026
Incoming Software Engineer Intern (Distributed Systems) *Washington D.C.*

State Farm May 2025 – August 2025
Software Engineer Intern *Dunwoody, GA*

- Migrated a high-throughput Spring Boot service to a Python/FastAPI backend on AWS Lambda + ECS (Fargate), reducing API latency by **40%** and serving **100K+** requests monthly
- Cut deployment time by **95%+** by building a GitLab CI/CD pipeline across Linux environments, automating **200+** unit & integration tests via Docker-isolated sandboxes

Georgia Tech - College of Computing August 2024 – Present
Instructor for Introduction to Operating Systems *Atlanta, GA*

- Designed and taught curriculum on Linux kernel internals: process scheduling, virtual memory, system calls, paging/TLB, and synchronization primitives (mutexes, semaphores, condition variables)
- Built autograding infrastructure executing **760K+ test runs**/semester, including C/C++ kernel-mode assignments testing multithreading, memory management, and IPC across **2,000+ students**
- Led weekly lecture with **100+** students covering: TCP/IP networking stack, socket programming, cache coherence protocols, computer architecture, and CPU pipeline hazards

Georgia Tech - Quantum Computing Lab November 2023 – Present
Research Assistant *Atlanta, GA*

- Created Quantum Playground, a browser-based quantum sandbox; deployed on Linux (AWS EC2) with Nginx reverse proxy, `systemd` service management, and kernel tuning to handle **20K+ concurrent requests**
- Built a Rust-based REPL shell via PyO3 with memory-efficient qubit encodings, leveraging Rust's ownership model to eliminate data races in multithreaded quantum algorithm execution

Projects

xv6 Kernel Extensions | *C, x86 Assembly, GDB, QEMU*

- Extended xv6 with CFS (red-black tree run queues, virtual runtime tracking) and copy-on-write fork, cutting memory footprint by **60%**; profiled with `perf` and debugged via GDB hardware watchpoints
- Implemented user-space threading via `clone()` syscall with custom stack allocation and mutex/condvar primitives built on atomic compare-and-swap

Checkpoint Autograder | *C/C++, Python, Docker, SSH, MIPS Assembly, Linux namespaces/cgroups*

- Built a sandboxed autograding platform for OS/systems assignments, including a GDB-style C++ debugger with breakpoints, watchpoints, and live register/memory inspection of student kernel code on simulated MIPS hardware
- Processed **200K+ submissions/year** across **2,000+ students**, cutting grading latency by **65%** via parallelized test execution and reducing manual TA grading time by **94%**
- Enforced isolation using Linux namespaces, cgroups, and seccomp-bpf syscall filtering per submission; integrated compiler toolchain with IRQ handlers and memory-protection mechanisms for low-level C/MIPS assignments

Technical Skills

Languages: C/C++, Rust, Python, Go, Java, x86 &RISC-V Assembly, TypeScript
Systems: Linux Kernel Internals, POSIX Threads, TCP/IP Socket Programming
Infrastructure: Docker (namespaces/cgroups), Linux cgroups/seccomp, Nginx, AWS EC2, Ansible, CI/CD
Concepts: Multithreading & Synchronization, Memory Hierarchy & Cache Optimization, Distributed Systems, SIMD/Vectorization, GPU (CUDA)